System Settings

What are System Settings?

System settings are the options that **configure how the computer hardware and operating system behave**. They control things like display, sound, network, power, date and time, user accounts and startup. They exist at two levels:

1. **Hardware level:** BIOS/UEFI (CMOS) setup
2. **Operating system level:** Windows Settings and Control Panel

1. BIOS/UEFI Setup (Hardware-Level Settings)

What is BIOS?

BIOS (Basic Input Output System) is firmware stored on a chip on the motherboard. It runs first when you switch on the computer, tests the hardware (**POST: Power On Self Test**), and then loads the operating system. **UEFI (Unified Extensible Firmware Interface)** is the modern replacement, with a mouse-driven graphical interface, faster boot and support for drives larger than 2 TB.

Its settings are saved in **CMOS memory**, which is kept alive by the **CMOS battery** (CR2032).

How to Enter BIOS Setup

Press a key repeatedly right after switching on the computer:

Brand	Common Key
Most desktops	**Del** or **F2**
Dell	F2 or F12
HP	F10 or Esc
Lenovo	F1 or F2
Asus	Del or F2
Acer	F2 or Del
Boot menu only	F12, F9 or Esc

On Windows 10/11 you can also go to **Settings → System → Recovery → Advanced startup → Restart now → Troubleshoot → UEFI Firmware Settings.**

Main BIOS/UEFI Settings

Section	Settings
Main / System Information	Date and time, BIOS version, CPU type, RAM size, serial number
Boot	Boot order (HDD, SSD, USB, DVD, network), fast boot, boot mode (UEFI or Legacy)
Advanced	CPU features (virtualization, hyper-threading), SATA mode (AHCI/RAID/IDE), USB and onboard device settings
Security	Supervisor/user password, **Secure Boot**, TPM
Power	Power-on after power failure, wake-on-LAN, ACPI sleep states
Overclocking / Performance	CPU multiplier, memory speed (XMP/EXPO profile), voltages, fan speed curves
Monitoring (H/W Monitor)	CPU temperature, fan speed, voltages
Save and Exit	Save changes, discard changes, **Load Optimized (Default) Settings**

Important BIOS Settings Explained

- **Boot order:** decides which device the computer starts from. Change it to boot from a USB drive when installing an OS.
- **SATA mode:** AHCI is recommended for modern drives. Changing it after installing Windows can cause a boot failure.
- **Secure Boot:** allows only trusted, signed operating systems to start (required for Windows 11).
- **TPM (Trusted Platform Module):** hardware security feature used for encryption (BitLocker); required for Windows 11.
- **Virtualization (Intel VT-x, AMD-V/SVM):** needed to run virtual machines and Android emulators.
- **XMP / EXPO:** one-click profile that runs RAM at its rated speed instead of the default slower speed.
- **BIOS password:** prevents others from changing settings or booting the PC.
- **Reset to defaults:** fixes problems caused by wrong settings.

Clearing/Resetting CMOS

If the BIOS is locked or the computer will not start after a wrong setting:

1. Switch off and unplug the computer.
2. Remove the CMOS battery for a minute, then reinstall it, **or** use the **CLR_CMOS jumper** on the motherboard.
3. Reconnect and start the computer. Settings return to defaults.

Updating BIOS (Flashing)

A BIOS update adds CPU support and fixes bugs. Download it from the motherboard manufacturer's site and use the built-in flash tool or Windows utility. **Never switch off the computer during an update**, because it can make the motherboard unusable.

2. Operating System Settings (Windows)

Ways to Open System Settings

- **Settings app:** press **Windows + I**
- **Control Panel:** Start → type "Control Panel" (older, but still used)
- **Run command:** press **Windows + R** and type a command (see the table below)
- **Right-click Start menu:** quick links to Device Manager, Disk Management and more

Windows Settings App: Main Categories

Category	What You Can Do
System	Display, sound, notifications, focus, power and battery, storage, multitasking, clipboard, remote desktop, About
Bluetooth and devices	Add Bluetooth devices, printers and scanners, mouse, touchpad, USB, AutoPlay
Network and internet	Wi-Fi, Ethernet, VPN, airplane mode, mobile hotspot, proxy, data usage
Personalization	Background, colors, themes, lock screen, taskbar, Start menu, fonts
Apps	Installed apps, default apps, startup apps
Accounts	Your info, sign-in options (PIN, password), family, email accounts
Time and language	Date and time, language, region, keyboard, typing
Gaming	Game mode, captures
Accessibility	Text size, magnifier, narrator, captions, high contrast
Privacy and security	Permissions (camera, microphone, location), Windows Security, device encryption
Windows Update	Check for and install updates, update history, pause updates

---

Key System Settings in Detail

A. Display Settings

Resolution, scale, refresh rate, orientation, brightness, night light, HDR and multiple displays (covered in the previous topic).

B. Sound Settings

- Choose the **output device** (speakers, headphones, HDMI) and **input device** (microphone)
- Adjust master volume and per-app volume (**Volume mixer**)
- Enable or disable devices and run the sound troubleshooter

C. Power and Battery Settings

- **Power modes:** Best power efficiency, Balanced, Best performance
- **Screen and sleep timeouts:** when the display turns off and the PC sleeps
- **Battery saver:** limits background activity at low charge
- **Lid and power button actions** (Control Panel → Power Options): sleep, hibernate or shut down

State	Meaning
Sleep	Low power; work stays in RAM; resumes quickly
Hibernate	Saves RAM contents to disk and powers off completely
Shut down	Closes everything and powers off
Restart	Closes everything and reboots
Fast startup	Hybrid of shutdown and hibernate for quicker boot

D. Storage Settings

- View disk usage by category
- **Storage Sense:** automatically deletes temporary files
- Manage drives, create and format volumes (also via **Disk Management**)
- Change where new apps and files are saved

E. Date and Time Settings

- Set time zone (India uses IST, UTC+5:30), automatic time sync and date/time format
- Wrong time often means the **CMOS battery** is weak

F. Language and Region

- Add display languages and keyboard layouts
- Set regional formats (date, currency, number)

G. Network Settings

- Connect to Wi-Fi or Ethernet
- View **IP address, DNS and MAC address** (ipconfig /all in Command Prompt)
- Set a static IP or DHCP (automatic)
- Configure VPN, proxy, hotspot and firewall

H. Accounts and Security Settings

- Create **administrator** and **standard** user accounts
- Set a **password, PIN, fingerprint or face sign-in**
- **User Account Control (UAC):** asks permission before system changes
- **Windows Security:** antivirus, firewall, device security, BitLocker encryption

I. Apps and Startup

- Uninstall or repair programs (Settings → Apps, or Control Panel → Programs and Features)
- Disable unnecessary **startup apps** to speed up boot (Task Manager → Startup tab)
- Set **default apps** for browser, mail, media

J. Windows Update

- Install security and feature updates and drivers
- Set **active hours** and pause updates temporarily
- Roll back a problematic update if needed

K. Privacy Settings

Control which apps may use the camera, microphone, location and diagnostic data.

---

3. Control Panel and System Tools

Important Control Panel Items

Item	**Purpose**
System	View processor, RAM, Windows edition, computer name
Device Manager	View and manage hardware and drivers
Programs and Features	Install, uninstall, repair software
Power Options	Create and edit power plans
Network and Sharing Center	Manage connections and adapters
User Accounts	Add or change users
Date and Time	Clock and time zone
Sound	Playback and recording devices
Mouse and Keyboard	Pointer speed, buttons, repeat rate
Printers / Devices and Printers	Add and manage printers
Windows Defender Firewall	Allow or block network traffic

Backup and Restore / File History	Backup files
Troubleshooting	Automatic fixes for common problems

Device Manager

Shows all hardware installed. You can:

- **Update driver, roll back driver, disable or uninstall** a device
- See problems marked by a **yellow triangle** (driver problem) or a **down-arrow** (disabled)

Open it with **Windows + X → Device Manager,** or type devmgmt.msc in Run.

Useful Run Commands

Command	**Opens**
control	Control Panel
msconfig	System Configuration (boot options, startup, services)
devmgmt.msc	Device Manager
diskmgmt.msc	Disk Management
services.msc	Services
taskmgr	Task Manager
regedit	Registry Editor (use with care)
ncpa.cpl	Network Connections
appwiz.cpl	Programs and Features
powercfg.cpl	Power Options
sysdm.cpl	System Properties
cmd	Command Prompt
msinfo32	System Information
dxdiag	DirectX and hardware diagnostics
winver	Windows version
mstsc	Remote Desktop

System Properties (sysdm.cpl)

Has tabs for:

- **Computer Name:** change the PC name and join a domain or workgroup
- **Hardware:** Device Manager
- **Advanced: performance settings** (visual effects, virtual memory/page file), user profiles, startup and recovery

- **System Protection: System Restore** points
- **Remote:** enable Remote Desktop

Task Manager (Ctrl + Shift + Esc)

- **Processes:** see CPU, memory, disk and network use by each app; end frozen apps
- **Performance:** live graphs for CPU, RAM, disk, Wi-Fi and GPU
- **Startup apps:** enable or disable programs that run at boot
- **Users and Services:** manage background tasks

System Configuration (msconfig)

Controls **Safe Mode boot**, boot options and which services start. Use it for troubleshooting.

---

4. Other Important System Configuration Topics

Virtual Memory (Page File)

Part of the disk that Windows uses as extra RAM when RAM is full. Adjust it at **System Properties → Advanced → Performance → Settings → Advanced → Virtual memory**. Letting Windows manage it automatically is usually best.

Drivers

Small programs that let the OS communicate with hardware. Keep graphics, chipset, network and audio drivers updated through Windows Update or the manufacturer's website.

System Restore and Recovery

- **Restore point:** a snapshot of system files and settings you can return to if something breaks
- **Reset this PC:** reinstalls Windows (keep or remove files)
- **Safe Mode:** starts Windows with minimal drivers for troubleshooting (hold Shift while clicking Restart, then Troubleshoot → Startup Settings)
- **Advanced startup / Windows Recovery Environment:** repair tools, command prompt, startup repair

Environment Variables

Named values used by the OS and programs (such as PATH, TEMP). Found under System Properties → Advanced → Environment Variables.

Registry

A database of system and application settings (regedit). Changing it incorrectly can damage Windows, so **back it up first**.

Disk Management

Initialize, partition, format and assign drive letters. Use diskmgmt.msc.

Security Best Practices

- Keep Windows and antivirus updated
- Use strong passwords and enable a BIOS password if needed
- Use a standard account for daily work
- Turn on the firewall and BitLocker (if available)
- Back up important data regularly

5. System Settings in Other Operating Systems

OS	**Where Settings Are Found**
Windows	Settings app (Win + I) and Control Panel
macOS	**System Settings** (Apple menu → System Settings)
Linux (Ubuntu)	Settings app; command-line tools such as nmcli, timedatectl, systemctl
Android	Settings app (gear icon)
iOS/iPadOS	Settings app